Third Examination 2025
Artificial Intelligence and Machine Learning

2025/01/02

1. Given a categorical dataset contained in the file named “DTclsn_trn.csv”, please apply classification and regression tree (CART) to train a classifier for the data with 5 classes. Note that the categorical data is a 7-dimensional data. There are 5, 5, 7, 5, 8, 4, and 6 categories in feature 1 to 7, respectively. The numbers in column 1 to 7 are categorical types of feature 1 to 7, respectively, and the classes the data belongs to are listed in the last column. Test your classifier on the testing dataset in the file named “DTclsn_tst.csv”. (a) Please write a report with the file name “report1.pdf” showing the process details you did. (b) Generate a file named “DTclsn_result.csv” containing the testing results based on the classifier you learned with CART. Each row of the data in the file “DTclsn_result.csv” should contain [testing data x], [class r]. (c) Explain the criteria you made in your source code in order to obtain the convergent solutions. In addition to the file “report1.pdf”, please also hand in the pdf file named “source1.pdf” containing your source code. Note that you can ONLY use the decision tree approach CART to learn and test your classifier. Any approach other than decision tree approach WILL NOT GET ANY SCORE at all. (25%)
2. Given a categorical dataset contained in the file named “DTregrs_trn.csv”, please apply classification and regression tree (CART) to train a regressor for the data. Note that the data for regression is a 6-dimensional input data associated with a 1-dimensional output data. Test your regressor on the testing dataset in the file named “DTregrs_tst.csv”. (a) Please write a report with the file name “report2.pdf” showing the process details you did. (b) Generate a file named “DTregrs_result.csv” containing the testing results based on the regressor you learned with CART. Each row of the data in the file “DTregrs_result.csv” should contain [testing data x], [regression output y]. (c) Explain the criteria you made in your source code in order to obtain the convergent solutions. In addition to the file “report2.pdf”, please also hand in the pdf file named “source2.pdf” containing your source code. Note that you can ONLY use the decision tree approach CART to learn and test your regressor. Any approach other than decision tree approach WILL NOT GET ANY SCORE at all. (25%)
3. Given a dataset contained in the file named “LDclsn_trn.csv”, please apply linear discrimination approach to train classifiers for the data. Test your classifiers on the testing dataset in the file named “LDclsn_tst.csv”. (a) Please write a report with the file name “report3.pdf” showing the process details you did. (b) Generate files

named “LDclsn_result.csv” containing the testing results based on the classifier you learned. Each row of the data in the files “LDclsn_result.csv” should contain [testing data x], [class r]. (c) Please show the discriminant function you have learned. (d) Explain the criteria you made in your source code in order to obtain the convergent solutions. In addition to the file “report3.pdf”, please also hand in the pdf file named “source3.pdf” containing your source code. Note that you can ONLY use linear discrimination approach to learn and test your classifier. Any approach other than linear discrimination approach WILL NOT GET ANY SCORE at all. (25%)

4. Given an 8-dimensional dataset in the file named “SVMclsn_trn.csv” for training, please train a classifier using support vector machine (SVM) with polynomial kernels to learn a classifier for the data. You may design your SVM with appropriate parameters and order for your polynomial kernels. Test your classifier on the testing dataset named “SVMclsn_tst.csv”. (a) Please write a report with the file name “report4.pdf” showing the process details you did. (b) Generate a file named “SVMclsn_result.csv” containing the testing results based on the testing data with learned classifier based on SVM. Each row of the data in the file “SVMclsn_result.csv” should contain [testing data x], [class r]. (c) Please show the polynomial kernels you have learned. (d) Explain the criteria you made in your source code in order to obtain the convergent solutions. In addition to the file “report4.pdf”, please also hand in the pdf file named “source4.pdf” containing your source code. Note that you can ONLY use SVM with polynomial kernels to learn and test your estimator. Any approach other than SVM with radial basis functions WILL NOT GET ANY SCORE at all. (25%)

You should upload all files to your account in Google Drive no later than 22:00 pm, Jan. 3, 2025. Any files submitted later than the due time will not be accepted.